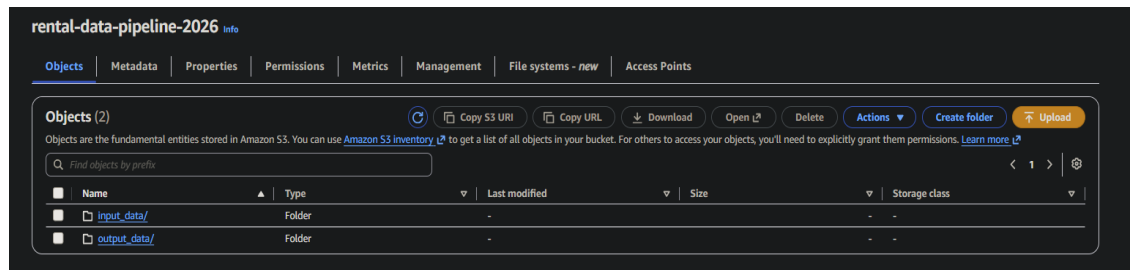


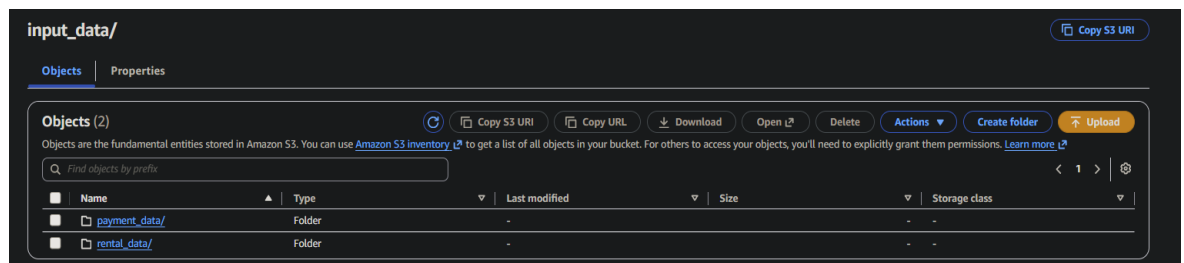
AWS observation

1.1

S3 is our raw data storage. We store CSV files here which trigger the pipeline. The folder structure separates raw input data from query results.(Picture 1,2)



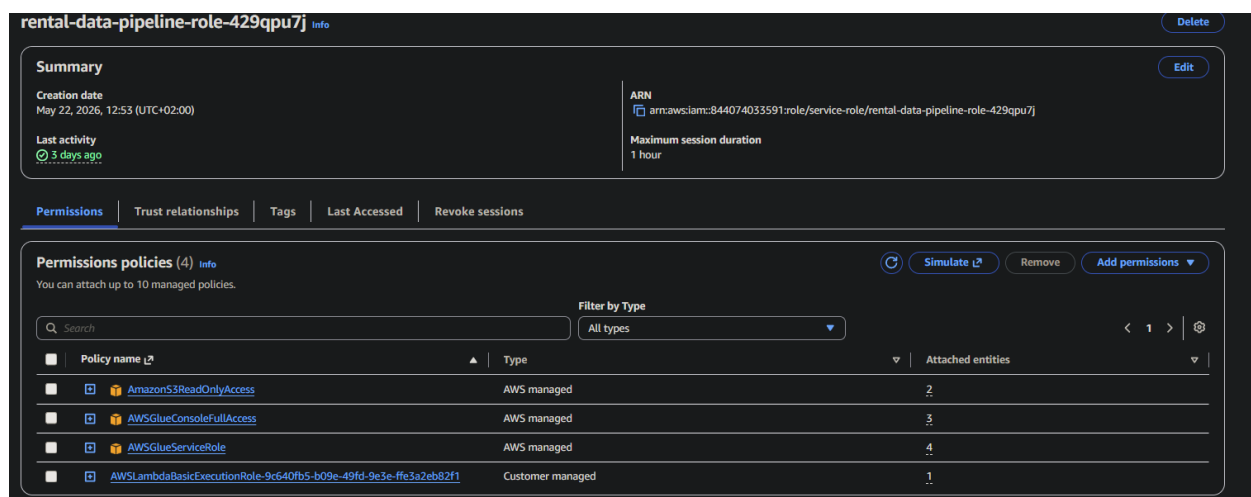
Picture -1



Picture – 2

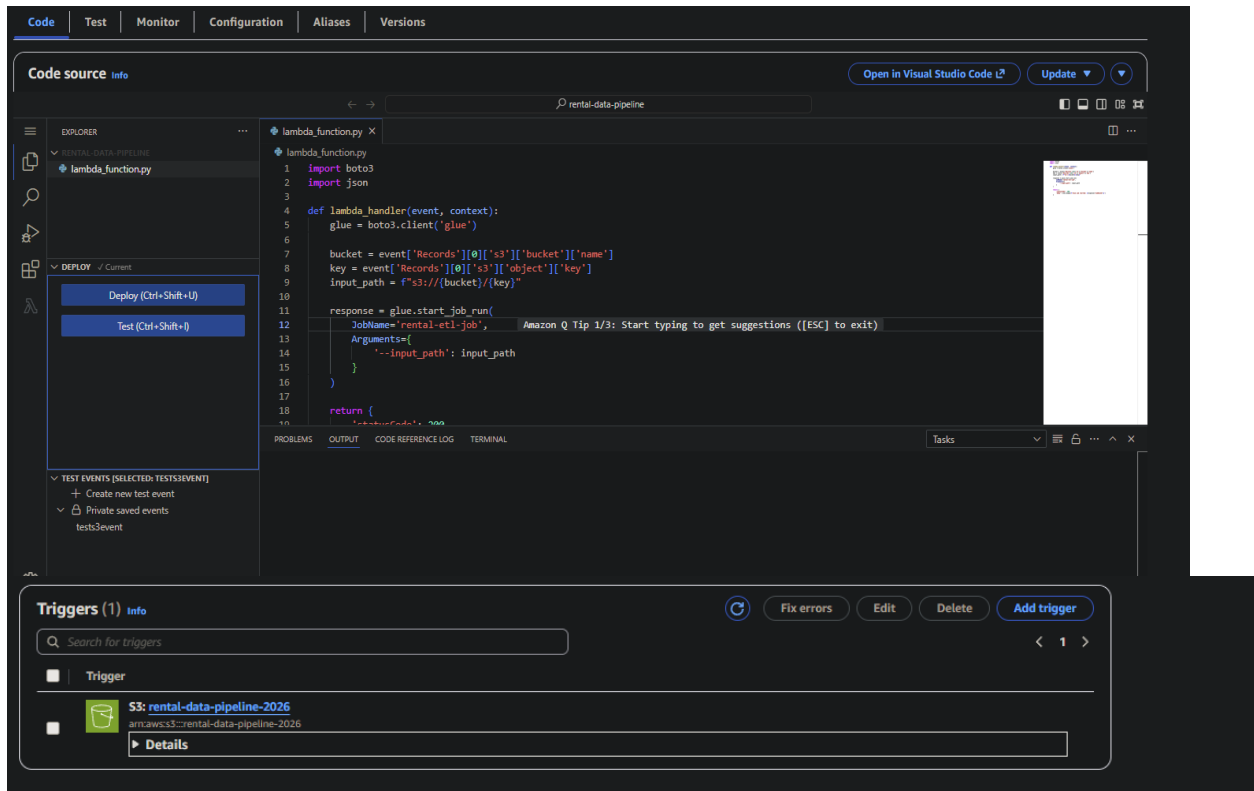
1.2

IAM controls which services can access each other. Without proper IAM roles, Lambda cannot trigger Glue and Glue cannot read from S3. Here we show that Lambda has all necessary permissions.(Picture – 3)



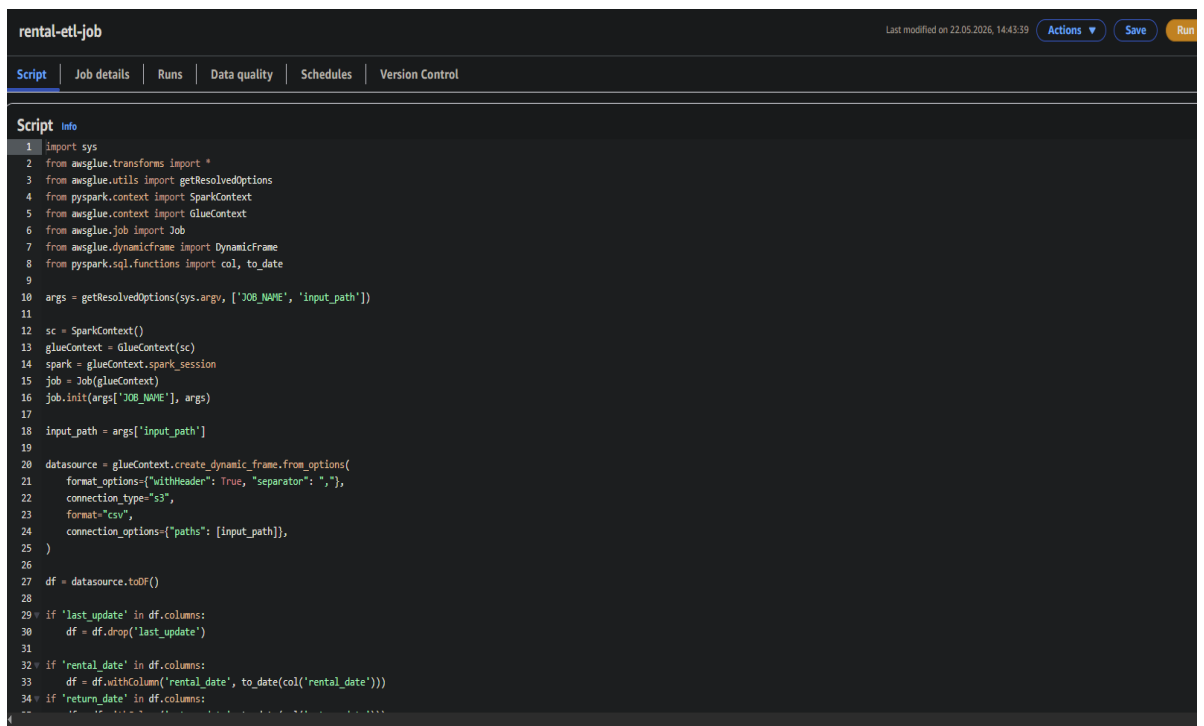
Picture – 3

1.3 Lambda is our event-driven trigger. It automatically detects when a new CSV file arrives in S3 and starts the Glue ETL job. No servers needed — it runs only when triggered.(picture - 4)



Picture – 4

1.4 Glue is our ETL engine. It reads raw CSV files from S3, cleans the data by dropping unnecessary columns and converting data types, then saves processed data back to S3 in Parquet format. Pictures (5,6,7)



Picture – 5

```

19
20 datasource = glueContext.create_dynamic_frame.from_options(
21     format_options={"withHeader": True, "separator": ","},
22     connection_type="s3",
23     format="csv",
24     connection_options={"paths": [input_path]},
25 )
26
27 df = datasource.toDF()
28
29 if 'last_update' in df.columns:
30     df = df.drop('last_update')
31
32 if 'rental_date' in df.columns:
33     df = df.withColumn('rental_date', to_date(col('rental_date')))
34 if 'return_date' in df.columns:
35     df = df.withColumn('return_date', to_date(col('return_date')))
36
37 for column in ['rental_id', 'inventory_id', 'customer_id', 'staff_id']:
38     if column in df.columns:
39         df = df.withColumn(column, col(column).cast('integer'))
40
41 output_path = "s3://rental-data-pipeline-2026/input_data/rental_data/processed/"
42
43 dynamic_frame = DynamicFrame.fromDF(df, glueContext, "dynamic_frame")
44
45 glueContext.write_dynamic_frame.from_options(
46     frame=dynamic_frame,
47     connection_type="s3",
48     format="parquet",
49     connection_options={"path": output_path}
50 )
51
52 job.commit()

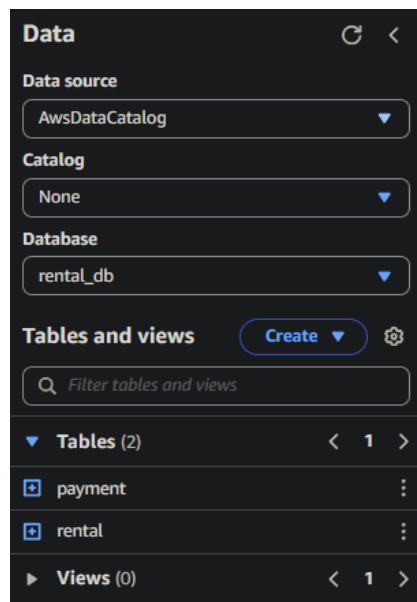
```

Picture – 6

Run status	Retries	Start time (Local)	End time (Local)	Duration	Capacity (DPU's)	Worker type	Glue version
● Succeeded	0	05/22/2026 15:24:48	05/22/2026 15:25:51	56 s	2 DPU's	G.1X	5.1
● Succeeded	0	05/22/2026 15:21:50	05/22/2026 15:22:59	1 m 2 s	2 DPU's	G.1X	5.1
● Succeeded	0	05/22/2026 14:51:03	05/22/2026 14:52:06	56 s	2 DPU's	G.1X	5.1
● Succeeded	0	05/22/2026 14:48:10	05/22/2026 14:49:13	56 s	2 DPU's	G.1X	5.1
● Succeeded	0	05/22/2026 14:44:12	05/22/2026 14:45:22	57 s	2 DPU's	G.1X	5.1

Picture - 7

1.5 Athena allows us to run SQL queries directly on files stored in S3 without moving data anywhere. We use it for analysis of rental and payment data.(Pictures 8,9,10,11)



Picture 8

Total payments by customer

Query 1

```
1 SELECT customer_id, round(SUM(amount),2) as total_payments
2 FROM rental_db.payment
3 GROUP BY customer_id
4 ORDER BY total_payments DESC
5 LIMIT 10;
```

SQL Ln 1, Col 41

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 95 ms Run time: 502 ms Data scanned: 879.81 KB

Results (10)

Search rows

#	customer_id	total_payments
1	526	221.55
2	148	216.54
3	144	195.58
4	178	194.61

Picture 9

All rentals joined with payments

Query 1

```
1 SELECT r.rental_id, r.rental_date, r.customer_id, p.amount, p.payment_date
2 FROM rental_db.rental r
3 JOIN rental_db.payment p ON r.rental_id = p.rental_id
4 LIMIT 20;
```

SQL Ln 4, Col 10

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 86 ms Run time: 856 ms Data scanned: 1.46 MB

Results (20)

Search rows

#	rental_id	rental_date	customer_id	amount	payment_date
1	2	2022-05-24 21:54:33.000000 +00:00	459	2.99	2022-02-21 17:38:11.389365 +00:00
2	3	2022-05-24 22:03:39.000000 +00:00	408	3.99	2022-03-23 15:44:32.990689 +00:00
3	4	2022-05-24 22:04:41.000000 +00:00	333	4.99	2022-04-12 07:42:01.166425 +00:00
4	5	2022-05-24 22:05:21.000000 +00:00	222	6.99	2022-04-03 11:47:13.012845 +00:00
5	6	2022-05-24 22:06:07.000000 +00:00	549	8.99	2022-04-10 11:06:45.478746 +00:00

Picture 10

Active rentals without return date



Query 1:

```
1 SELECT * FROM rental_db.rental
2 WHERE return_date = ''
3 OR return_date IS NULL
4 LIMIT 20;
```

SQL Ln 4, Col 10

Run again Explain L2 Cancel Clear Create

Query results Query stats

Completed Time in queue: 130 ms Run time: 376 ms Data scanned: 1.21 MB

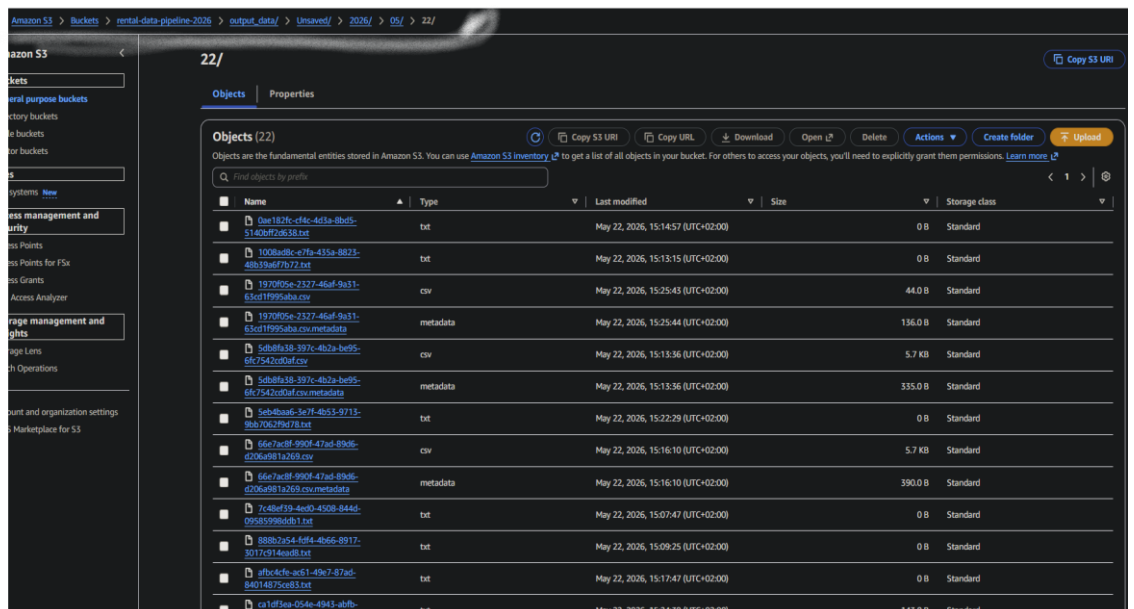
Results (20)

Search rows

#	rental_id	rental_date	inventory_id	customer_id	return_date	staff_id	last_update
1	11496	2022-02-14 15:16:03.000000 +00:00	2047	155		1	2022-02-16 02:30:53.000000 +00:00
2	11541	2022-02-14 15:16:03.000000 +00:00	2026	335		1	2022-02-16 02:30:53.000000 +00:00
3	12101	2022-02-14 15:16:03.000000 +00:00	1556	479		1	2022-02-16 02:30:53.000000 +00:00

Picture 11

The Folder with our query results



Amazon S3 Buckets rental-data-pipeline-2026 output_data/ Unsaved/ 2026/ 05/ 22/ Copy S3 URI

Objects Properties

Objects (22)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
5ae182fc-cf4c-4d3a-8bd5-514260726338.txt	txt	May 22, 2026, 15:14:57 (UTC+02:00)	0 B	Standard
1008a8dc-e7fa-435a-8823-48639aef7b72.txt	txt	May 22, 2026, 15:13:15 (UTC+02:00)	0 B	Standard
1970f05e-2327-46af-9a31-63cd1f995d8a.csv	csv	May 22, 2026, 15:25:43 (UTC+02:00)	44.0 B	Standard
1970f05e-2327-46af-9a31-63cd1f995d8a.csv.metadata	metadata	May 22, 2026, 15:25:44 (UTC+02:00)	136.0 B	Standard
5db8ffa38-397c-462a-be95-6fc7542c03af.csv	csv	May 22, 2026, 15:13:36 (UTC+02:00)	5.7 KB	Standard
5db8ffa38-397c-462a-be95-6fc7542c03af.csv.metadata	metadata	May 22, 2026, 15:13:36 (UTC+02:00)	335.0 B	Standard
5eb4baaf-5e7f-4b53-9713-96b706279d78.txt	txt	May 22, 2026, 15:22:29 (UTC+02:00)	0 B	Standard
6fe7ac8f-9908-47ad-89dc-c706a981a269.csv	csv	May 22, 2026, 15:16:10 (UTC+02:00)	5.7 KB	Standard
6fe7ac8f-9908-47ad-89dc-c706a981a269.csv.metadata	metadata	May 22, 2026, 15:16:10 (UTC+02:00)	390.0 B	Standard
7c48ef39-4e0b-4308-844d-09585998d081.txt	txt	May 22, 2026, 15:07:47 (UTC+02:00)	0 B	Standard
888b2a54-fdf4-4b66-8917-3017c914a9d8.txt	txt	May 22, 2026, 15:09:25 (UTC+02:00)	0 B	Standard
afbc4cfe-ac51-49e7-87ad-84014075c983.txt	txt	May 22, 2026, 15:17:47 (UTC+02:00)	0 B	Standard
ca1df3ea-054e-4943-abfb-05baf6e7a269.txt	txt	May 22, 2026, 15:24:39 (UTC+02:00)	143.0 B	Standard